A Systematic Multi-Asset Strategy: From Data to Backtest

Jean Philippe N'Dri • 6 October 2026 • code: `multiasset-strat`, built on `portopt`

Abstract. Each month the strategy selects 10 lines out of 132 assets (US stocks, ETFs, bonds, commodities) using price signals, weights them by risk budgeting, and limits rotations to account for a fixed fee per order. Over 2018–2026, a period not used to choose the signals, the net Sharpe ratio is 1.15 against 0.82 for a 60/40 portfolio. This result is overstated by an uncorrected survivorship bias.

1 Data and universe

Adjusted closing prices (dividends reinvested) come from Yahoo Finance from 2008 onwards and are converted into daily simple returns $r_{i,t} = P_{i,t}/P_{i,t-1} - 1$. The universe has four sleeves and one safe asset:

- **Stocks** (96): large caps from the S&P 500 and the Nasdaq 100;
- **ETFs** (21): indices, factors, themes, multi-asset;
- **Bonds** (8) and **commodities** (7);
- **Cash**: `SHY`, which receives the capital of empty slots.

An asset is eligible at date t only if it has 252 consecutive returns before t, so late listings enter the universe naturally. The ETFs are US proxies; live execution goes through UCITS equivalents.

2 Signals

All signals are computed on the window $[t-252,\, t)$, with no information after t. Thirteen price signals taken from the literature were tested one by one, per sleeve, on 2009–2017, then checked once on 2018–2026. Five are kept.

Trend (stocks, ETFs, commodities):

$$\mathrm{mom}_{12\text{-}1} = \frac{P_{t-21}}{P_{t-252}} - 1, \qquad \mathrm{mom}_{6\text{-}1} = \frac{P_{t-21}}{P_{t-126}} - 1,$$

$$\mathrm{ma}_{200} = \frac{P_t}{\mathrm{SMA}_{200}} - 1, \quad \mathrm{ma}_{50/200} = \frac{\mathrm{SMA}_{50}}{\mathrm{SMA}_{200}} - 1.$$

Mean reversion (bonds):

$$\mathrm{boll}_{20} = -\,\frac{P_t - \mathrm{SMA}_{20}}{\sigma_{20}(P)}.$$

Each signal $x_{i,k}$ is winsorised at 1%/99% and standardised *within its sleeve* g; the score is the average of the sleeve's K_g signals:

$$z_{i,k} = \frac{x_{i,k} - \mu_k^{(g)}}{\sigma_k^{(g)}}, \qquad S_i = \frac{1}{K_g}\sum_{k=1}^{K_g} z_{i,k}.$$

Standardising per sleeve avoids comparing the momentum of a stock with that of a bond, whose dispersions have nothing in common. An **absolute filter** is added: an asset can be bought only if $P_t > \mathrm{SMA}_{200}$.

Low volatility, part of the first version of the score, was removed: its information coefficient is negative for stocks and ETFs in both sub-periods.

3 Selection

The portfolio holds 5 stocks, 2 ETFs, 2 bonds and 1 commodity. In each sleeve the best scores are picked among the assets passing the filter. Three rules limit rotations:

- **hysteresis**: a held line leaves only if it breaks its trend or drops out of the top 40% of its sleeve;
- **at most two exits** per rebalancing, broken trends first;
- a slot with no candidate is parked in `SHY`.

4 Allocation

The weights of the n selected lines solve a risk budgeting problem. With $\boldsymbol{\Sigma}$ the Ledoit–Wolf covariance estimated on the window, the relative contribution of line i is

$$\mathrm{RC}_i = \frac{w_i\,(\boldsymbol{\Sigma}\boldsymbol{w})_i}{\boldsymbol{w}^\top\boldsymbol{\Sigma}\boldsymbol{w}}, \qquad \sum_i \mathrm{RC}_i = 1,$$

and we impose $\mathrm{RC}_i = b_i$, where the sleeve budgets are 40% for stocks, 24% for ETFs, 24% for bonds and 12% for commodities, split equally between the lines of a sleeve. The solution is obtained by coordinate descent on Spinu's convex formulation,

$$\boldsymbol{y}^\star = \arg\min_{\boldsymbol{y}>0}\ \tfrac{1}{2}\,\boldsymbol{y}^\top\boldsymbol{\Sigma}\boldsymbol{y} - \sum_i b_i \ln y_i, \qquad \boldsymbol{w} = \frac{\boldsymbol{y}^\star}{\mathbf{1}^\top\boldsymbol{y}^\star}.$$

Weights are then bounded ($5\% \le w_i \le 20\%$, with a 3% floor for a stock). Since these bounds distort the budgets, the stock budget is re-imposed as a ceiling by bisection on a scaling factor applied to the sleeve.

5 Turnover constraints

Let $\boldsymbol{w}^-$ be the portfolio held (drifted weights) and $\boldsymbol{w}^\star$ the target. Because fees are fixed per order, cost depends on the *number* of orders, not on the amount traded. Execution therefore applies:

- a **no-trade band**: no order if $|w_i^\star - w_i^-| < 2$ points;
- a **cap** $\tau_t = \sum_i |w_{i,t} - w_{i,t}^-| \le 40\%$, giving up the smallest adjustments first.

Weight bounds take precedence over the cap, which stays soft for the last rotation.

6 Backtest

The backtest is walk-forward: at each date t of the grid (every 21 trading days), weights are computed on $r_{[t-252,\,t)}$ and applied from r_t onwards, then drift freely until the next rebalancing. The cost paid at t is

$$c_t = s\,\tau_t + \frac{f\,n_t^{\mathrm{orders}}}{\mathrm{NAV}_t},$$

with $s = 5$ bps, $f = 1$ EUR per order and $\mathrm{NAV}_0 = 10{,}317$ EUR. A fixed fee therefore weighs more on a small account. Two benchmarks bear the same fees: a 60/40 (`SPY`/`IEF`) and a risk parity on 8 ETFs with no signal.

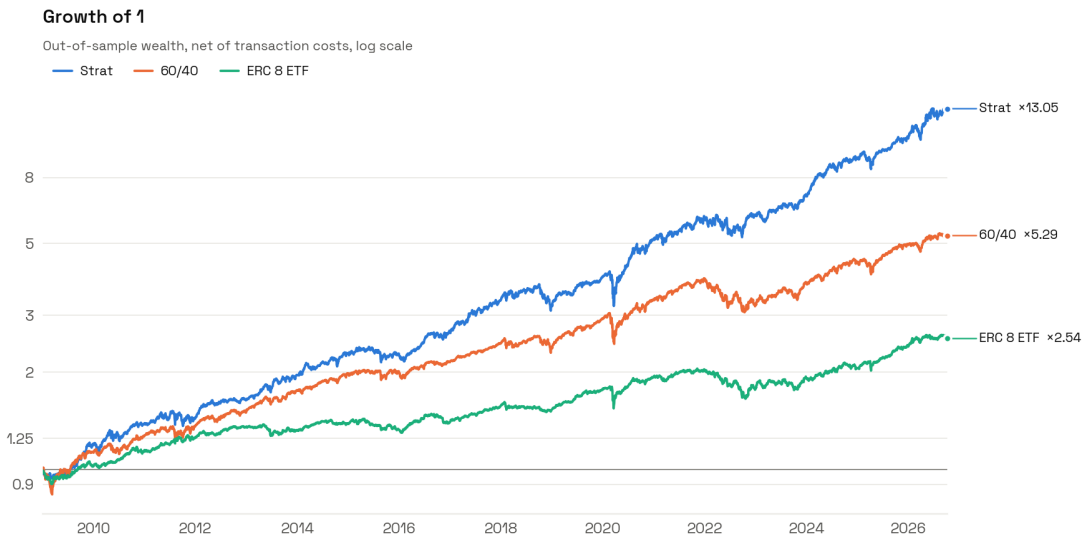


Figure 1: Growth of 1 invested, net of fees, log scale.

7 Results

Net of fees	Strategy	60/40	ERC 8 ETFs
2009–2026			
Annualised return	15.6%	9.9%	5.4%
Volatility	12.4%	10.3%	7.2%
Sharpe	1.23	0.96	0.76
Maximum drawdown	−21.9%	−21.5%	−19.4%
2018–2026, out of sample			
Annualised return	16.9%	9.2%	5.6%
Volatility	14.5%	11.5%	8.1%
Sharpe	1.15	0.82	0.71

Table 1: Performance net of fees. Signals were chosen on 2009–2017.

Over 2018–2026, the gap with the 60/40 is 7.2 points per year ($t = 2.4$). The strategy sends 4.3 orders per rebalancing on average, for 0.4% of annual fees and a turnover of 387% per year.

Risk checks. The portfolio never exceeds 10 lines or 21.7% per line. The stock sleeve carries 44% of risk on average (target 40%, maximum 59% between two rebalancings). Ex-ante variance splits into 62% market beta and 26% specific risk, of which 12% comes from stocks. The worst daily loss caused by a single line is 2.3 NAV points. The turnover cap is exceeded in 16.5% of rebalancings.

8 Limitations

- **Survivorship bias.** The stocks are today's large caps. A trend signal applied to survivors is flattered, by an amount that has not been quantified.
- **Weak statistical evidence.** No single signal survives the Benjamini–Hochberg correction over the 52 tests (best q-value: 0.46). The effect on stocks lives in the tail: the top 5 outperforms while the rank IC is zero.
- **Sector concentration.** There is no sector constraint: on 5 October 2026, the five selected stocks are all semiconductors.
- **Currency.** The backtest is in dollars, with no EUR/USD effect.